Algebra 1 Name____________________
Unit 9 Date_________________
Lesson 10 Practice Period_________

1. Use the quadratic formula to solve $x^2 - 14x + 66 = 4x$.

2. Solve the equation, $4x^2 + 32x + 60 = 0$, by factoring.

3. Solve the equation, $x^2 - 16x = -39$, by completing the square.

4. Solve the equation, $2x^2 - 2 = 3x$, with the method of your choice.

The equation $H(x) = -16x^2 + 64x + 192$ models the height, H, in feet, of a toy rocket after x seconds that it is launched from the top of a 16-story building.

5. Solve $-16x^2 + 64x + 192 = 0$ by factoring.

6. Interpret the solutions and determine if the solutions are viable.

A statistician for an outdoor company analyzes data for the years 2007 to 2014 to help inform the company about trends. The function $R(x) = -14x^2 + 112x + 776$ models the number of people ages 6 to 17, in tens of thousands, R, who listed running as a hobby for x years since 2007.

7. Solve $0 = -14x^2 + 112x + 776$ using the quadratic formula.

8. Interpret the solutions and determine if the solutions are viable.

Several investors provided funds for a startup company to begin production of a new product. The company's revenue, R, in thousands of dollars, for the first year is modeled by the function $R(x) = 4x^2 - 23x + 30$, where x is the number of months since production started.

9. Solve $0 = 4x^2 - 23x + 30$ using the method of your choice.

10. Interpret the solutions and determine if the solutions are viable.